

Mathematics - Probability and Statistics

Thomas de Jaeger
thomas.de-jaeger@epita.fr

Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Descriptive Statistics

Session 3

Descriptive Statistics

Session 3

Objectives:

- Explain the role of descriptive statistics in summarizing data.
- Define and compute mean, median, variance, and standard deviation.
- Interpret these measures in real-life examples.
- Distinguish when to use mean vs. median.
- Apply descriptive statistics using Python

Introduction: Why Descriptive Statistics?

Any time that you get a new data, one of the first tasks that you have to do summarize and simplify raw data (the other is....basic graphic representation).

This is what we called **Descriptive statistics**

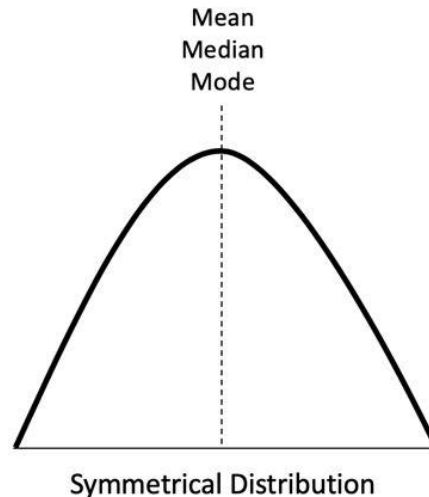
- Raw datasets can contain hundreds or thousands of values, which are hard to interpret directly.
- By reducing data to just a few key numbers, we make it easier to compare groups, detect unusual observations, and describe patterns.
- Two key questions:
 - Where is the **center** of the data (the “typical” value)?
 - How **spread out** are the values around that center?

Central Tendency: Finding a Typical Value

Mean: average value computed by adding up all values in a dataset and then dividing the sum by the number of values.

Median: The middle value in a set of ordered data.

Mode: The most frequent value in the dataset.



Central Tendency: The mean

The **mean** is the most common measure of central tendency. **It is very sensible to outliers.**

We measure, N times a variable X.

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_{N-1} + X_N}{N}$$

Ex: Scores [60, 70, 75, 80, 90] → $\langle X \rangle = (60 + 70 + 75 + 80 + 90) / 5 = 75$

This formula is entirely correct, but it's terribly long. We can use the compact formula:

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$$

In python

Import numpy as np

Mean_X=np.mean(X) where X=[60,70,75,80,90]

Central Tendency: The weighted mean

Sometimes not all observations count equally, so have measure the weighted average!!!

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Where w_i is the weight!!

Ex: Your final grade!! 65% final exam, 35% mid term. If you obtain 12/20 and 9/20 -> $(12*0.35+9*0.65)/(0.35+0.65)=10.05$

In python

```
weighted_mean = np.average(values, weights=weights)
```


Central Tendency: The median

The **median** is the central value when data are sorted.

- If n is odd $\rightarrow$ the exact middle observation.
- If n is even $\rightarrow$ the average of the two middle values.

Resistant to **outliers** and skewed distributions.

Ex: Scores [60, 70, 75, 80, 90] $\rightarrow$ median_X=**75**

In Python:

```
Median_X=np.median(X)
```

Mean or Median

Calculating means and medians is only the beginning; it's equally important to understand what each reveals about the data and when it is most appropriate to use them.

- In a symmetric distribution (bell-shaped), mean $\approx$ median.
- In a skewed distribution, the mean is pulled toward the tail.

Best practice: report both, but interpret carefully.

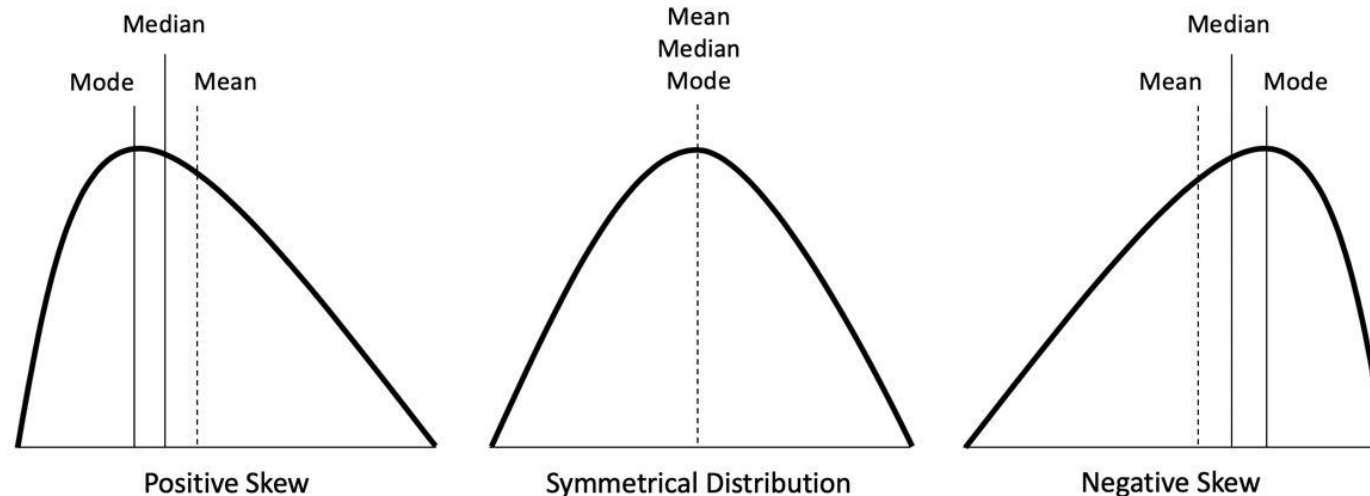
If you want to study the household income in a country, what will you choose?

Central Tendency: Finding a Typical Value

Mean: average value computed by adding up all values in a dataset and then dividing the sum by the number of values.

Median: The middle value in a set of ordered data.

Mode: The most frequent value in the dataset.



When to use what?

Nominal data: For categories with no order or ranking, **MODE**

Ordinal data: For categories that have a meaningful order, but distances between them are not measurable, **MODE or MEDIAN**

Numeric data: MEDIAN or MEAN IF DATA ARE SYMETRIC

You can apply Hildebrand criterion

$H = (\text{mean} - \text{median}) / \text{std}$, if $|H| < 0.20$, data are symetric

When to use what?

- T-shirt size of students (XL, L, M, S)
- Salary of Real Madrid players
- Restaurant ranking from best, average, worst

When to use what?

- T-shirt size of students (XL, L, M, S): **Ordinal data so MODE or MEDIAN is the data is odd**
- Salary of Real Madrid players: **Numerical data. Mbappé earns more than Asensio so there are outliers. Use MEDIAN**
- Restaurant ranking from best, average, worst: **Ordinal data so MODE or MEDIAN is the data is odd**

Mean, median mode

You have a list with the following values:

[1, 2, 2, 2, 2, 2, 3, 4, 6, 7, 8, 9, 9]

- What is the mean?
- What is the median?
- What is the mode?

How do these values change if we replace the last 9 with the number 100?

Mean, median mode

You have a list with the following values:

[1, 2, 2, 2, 2, 2, 3, 4, 6, 7, 8, 9, 9]

- What is the mean? **4.38**
- What is the median? **3**
- What is the mode? **2**

How do these values change if we replace the last 9 with the number 100? **Mean change to 11.38, mode and median the same**

Why Variability Matters

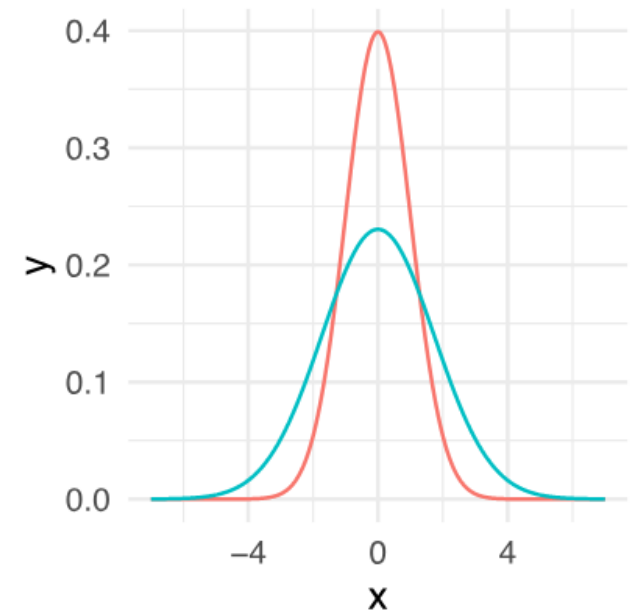
Mean/Median alone is not enough: two datasets can have the same mean but very different spreads.

Consistency vs unpredictability: Small variability = stable data, large variability = scattered data.

Detecting anomalies: Outliers and errors often appear through unusual variability.

Decision-making: Variability matters in finance (risk), science (measurement error), industry (quality control).

Foundation for statistics: Standard deviation, regression, and hypothesis testing rely on understanding variability.



Spread: MAD

First idea for spread: average how far values are from the mean.

$$MAD = \frac{1}{n} \sum |x_i - \bar{x}|$$

- Easy to interpret: It tells us the average distance from the mean in the same units as the data.

Student	Score (Xi)	Deviation (xi - <x>)	Absolute Deviation (xi - <x>)
1	45	-9	9
2	55	+1	1
3	65	+11	11
4	70	+16	16
5	35	-19	19
Total	270	0	56

$$MAD = 56/5 = 11.2$$

Spread: Variance

Variance quantifies how much values **differ from the mean on average**.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- x_i = the i -th value in the **sample**
- $\bar{x}$ = the **sample mean**
- n = number of observations in the sample
- s^2 = sample variance

Key properties:

- Always non-negative (squares can't be negative).
- Larger when data points are further from the mean.
- Units are **squared** (e.g., cm^2 , €^2).
- Dividing by $n-1$ (instead of n) corrects for bias → called **Bessel's correction**.

When we calculate variance **from a sample**, one degree of freedom is already used when measuring the mean. So the sample variance would underestimate the true population variance if we divided by n . Dividing by $n-1$ fixes this bias, making variance an unbiased estimator of the population variance.

Spread: Variance

Year	Departures (millions)	Acci- dents	Fatal- ities
1996	7.9	3	342
1997	9.9	3	3
1998	10.5	1	1
1999	10.9	2	12
2000	11.1	2	89
2001	10.6	6	531
2002	10.3	0	0
2003	10.2	2	22
2004	10.8	1	13
2005	10.9	3	22
2006	11.2	2	50

$$\bar{x} = \frac{\sum x_i}{n} = \frac{342 + 3 + 1 + 12 + 89 + 531 + 0 + 22 + 13 + 22 + 50}{11}$$

$$\bar{x} = \frac{1085}{11} \approx 98.64$$

$$\sum (x_i - \bar{x})^2 \approx 303,617$$

$$s^2 = \frac{303,617}{10} \approx 30,361.7$$

WHAT DOES THAT MEAN???

Spread: Variance

Year	Departures (millions)	Acci- dents	Fatal- ities
1996	7.9	3	342
1997	9.9	3	3
1998	10.5	1	1
1999	10.9	2	12
2000	11.1	2	89
2001	10.6	6	531
2002	10.3	0	0
2003	10.2	2	22
2004	10.8	1	13
2005	10.9	3	22
2006	11.2	2	50

$$\bar{x} = \frac{\sum x_i}{n} = \frac{342 + 3 + 1 + 12 + 89 + 531 + 0 + 22 + 13 + 22 + 50}{11}$$

$$\bar{x} = \frac{1085}{11} \approx 98.64$$

$$\sum (x_i - \bar{x})^2 \approx 303,617$$

$$s^2 = \frac{303,617}{10} \approx 30,361.7$$

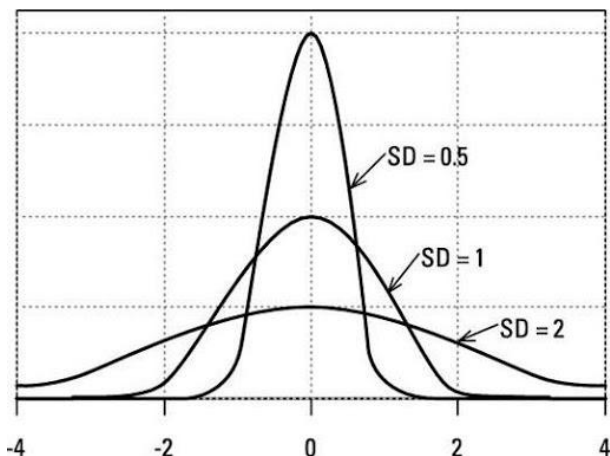
WHAT DOES THAT MEAN???

NEED TO INTRODUCE THE STANDARD DEVIATION

Standard deviation

Standard deviation: The square root of the variance. This measure is expressed in **the same units as the data itself**. It is also called the Root Mean Squared Deviation (RMSD).

The lower the standard deviation, the more the data is gathered around the mean. The larger the standard deviation, the more dispersed and far from the average the data is.



- Population standard deviation:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2}$$

- X_i : values in the population
- μ : population mean
- N : population size

- Sample standard deviation:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- x_i : observed values in the sample
- $\bar{x}$: sample mean
- n : sample size

Standard deviation: population or sample

- William went fishing ten times, and we know the number of fish he caught on each trip:

4, 6, 9, 5, 7, 5, 8, 3, 6, 7

We are asked to find the variance and standard deviation.

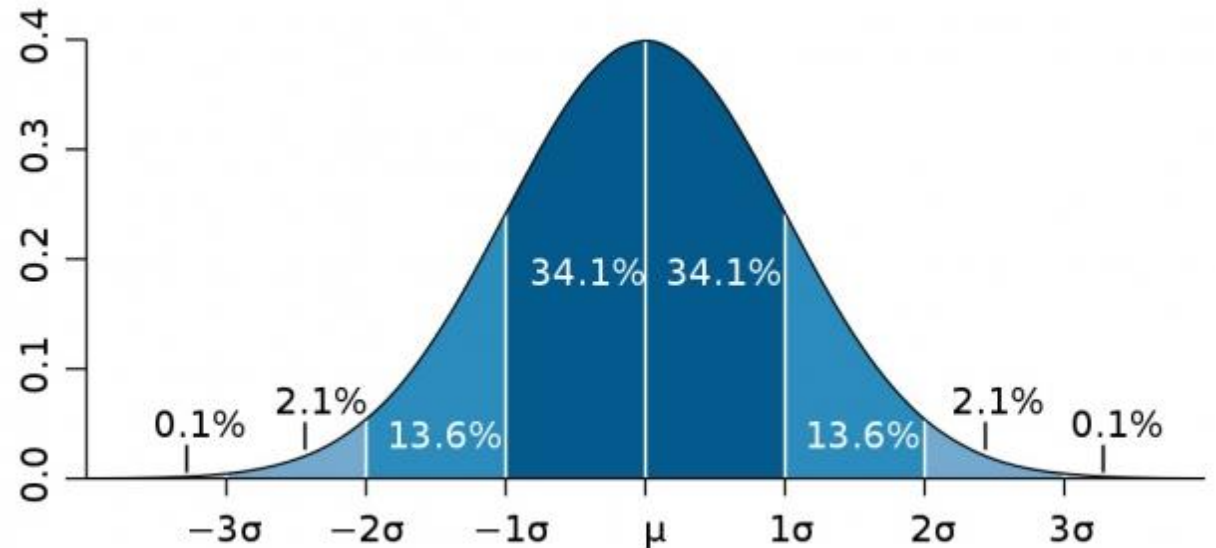
- Population formula (3,1.73)
- Five students are randomly selected from a math class and asked about their last exam scores. Their scores are: 92, 79, 85, 89, 80. Find the variance and standard deviation.
 - Sample formula (31.5,5.61)
- The following data represents the number of free throws James made in seven attempts: 40, 37, 42, 38, 46, 37, 47. Find the variance and standard deviation.
 - Population formula (14.86,3.85)

Standard deviation

EMPIRICAL RULE

When the data follow a bell-shaped distribution (Normal/Gaussian):

- 68% of the data values lie within one standard deviation: $\mu \pm 1 \sigma$
- 95% of the data values lie within two standard deviations: $\mu \pm 2 \sigma$
- 99.7% of the data values lie within three standard deviations: $\mu \pm 3 \sigma$



Standard deviation

Example:

You want to invest money in stock market. He discovers the following standard deviation of
One year of daily stock closing prices

Theocorp: std = 1.05 €

Adelecorp: std = 8.56 €

Where should he put his money?

Standard deviation

Example:

It depends!!

If you want a stable long-term investment, theocorp is better

However if you want to make a quick profit and want to take risk, adelecorp is better

Attention

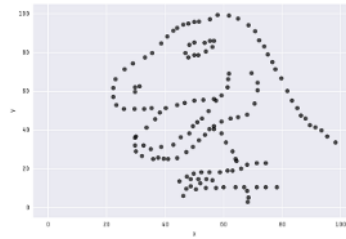
Statistics alone can be misleading.

Two datasets can have the same mean, variance, and correlation but tell very different stories.

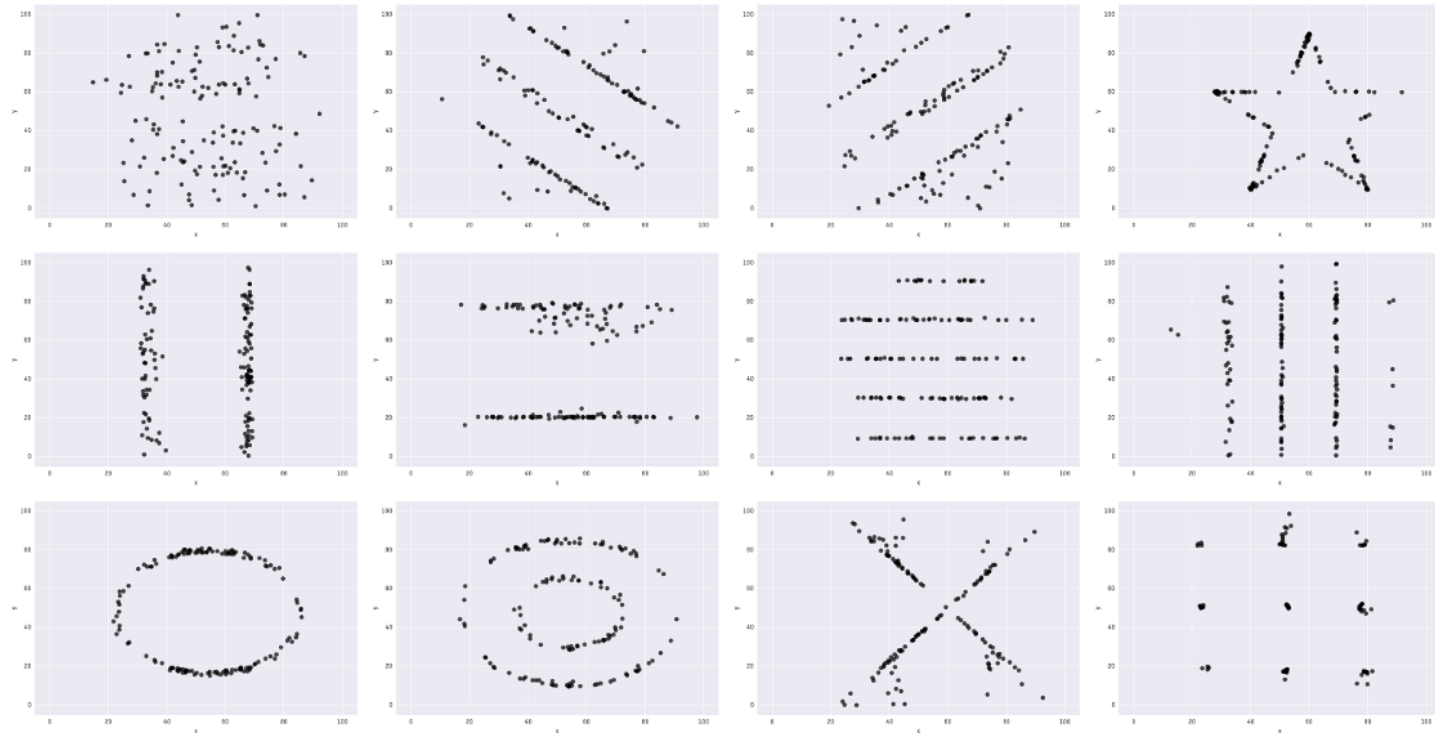
Visualization is essential.

Plotting the data reveals structure, outliers, or patterns that summary numbers hide.

Real-world lesson: **Never rely only on summary statistics. Always look at the raw data.**



X Mean: 54.26
Y Mean: 47.83
X SD : 16.76
Y SD : 26.93
Corr. : -0.06



Exercices

The following are the percentages of ash content in 12 samples of coal found in close proximity:

9.2, 14.1, 9.8, 12.4, 16.0, 12.6, 22.7, 18.9, 21.0, 14.5, 20.4, 16.9

(a) sample mean

(b) sample standard deviation of these percentage

Exercices

The following are the percentages of ash content in 12 samples of coal found in close proximity:

9.2, 14.1, 9.8, 12.4, 16.0, 12.6, 22.7, 18.9, 21.0, 14.5, 20.4, 16.9

(a) sample mean

$$\begin{aligned}\langle x \rangle &= (9.2 + 14.1 + 9.8 + 12.4 + 16.0 + 12.6 + 22.7 + 18.9 + 21.0 + 14.5 + 20.4 + 16.9) / 12 \\ &= 188.5 / 12 = 15.71\%\end{aligned}$$

$$(b) s^2 = 1 / (12 - 1) * [(9.2 - 15.71)^2 + (14.1 - 15.71)^2 + \dots + (16.9 - 15.71)^2]$$

$$\rightarrow s = 4.40\%$$

Exercices

The sample mean and sample variance of five data values are, respectively,
 $\langle x \rangle = 104$ and $s^2 = 16$.

If three of the data values are **102, 100, 105**, what are the **other two data values**?

Exercices

$$x=[102,100,105,x_4,x_5]$$

$$\langle x \rangle = 104 = (102 + 100 + 105 + x_4 + x_5) / 5$$

$$\rightarrow 520 = 307 + x_4 + x_5$$

$$\rightarrow x_4 + x_5 = 213$$

$$s^2 = 16 = 1/4 * [(102 - 104)^2 + (100 - 104)^2 + (105 - 104)^2 + (x_4 - 104)^2 + (x_5 - 104)^2]$$

$$\rightarrow (x_4 - 104)^2 + (x_5 - 104)^2 = 43$$

$\rightarrow$ Solve both equations and you should find:

$$x_4 = 102.6$$

$$x_5 = 110.4$$

Exercices

TOTAL GRADE:

- 15% quizzes
- 30% mid exam
- 50% final exam
- 5% participation

A student obtained

- 12/20, 15/20, 5/20 at the quizzes
- 8/20 mid term
- 20/20 participation

To pass the class, the student needs a total grade of 10/20.

What is the minimal final exam grade needed?

Exercices

Average quizzes $32/3=10.7$

$$10 = (10.7 * 0.15 + 0.3 * 8 + 20 * 0.05 + X * 0.5) / (0.15 + 0.3 + 0.5 + 0.05)$$

$$10 = 5.05 + 0.5X$$

$$X = (10 - 5.05) / 0.5$$

$$\mathbf{X=9.9}$$

Exercices

The exam scores of 8 students are: [55 , 60 , 62 , 65 , 65 , 70 , 72 , 100]

Compute mean, median, MAD, variance, and SD.

Interpret each measure in words.

If you were the professor, which measure would you use to report the “typical” grade and the “spread”? Why?

Exercices

The exam scores of 8 students are: [55 , 60 , 62 , 65 , 65 , 70 , 72 , 100]

Compute mean, median, MAD, variance, and SD.

Interpret each measure in words.

If you were the professor, which measure would you use to report the “typical” grade and the “spread”? Why?

- **Mean: 68.63**
- **Median: 65**
- **MAD (Mean Absolute Deviation): 9.03**
- **Variance: 189.70**
- **Standard Deviation: 13.77**
- **Median + MAD is more honest and robust for this dataset.**

Exercices

You are a researcher combining two surveys:

- **Survey A: 100 respondents, mean age = 30 years, SD = 5.**
- **Survey B: 40 respondents, mean age = 45 years, SD = 6.**

Compute the combined mean age of all 140 respondents.

Exercices

You are a researcher combining two surveys:

- **Survey A: 100 respondents, mean age = 30 years, SD = 5.**
- **Survey B: 40 respondents, mean age = 45 years, SD = 6.**

Compute the combined mean age of all 140 respondents.

Average=34.2857 years

Exercices

- Find the sample mean of the number of yearly airline accidents.
- Find the sample median of the number of yearly airline accidents.
- Find the sample mode of the number of yearly airline accidents.
- Find the sample standard deviation of the number of yearly airline accidents.

U.S. Airline Safety, Scheduled Commercial Carriers, 1985–2006

Year	Departures (millions)	Acci- dents	Fatal- ities	Year	Departures (millions)	Acci- dents	Fatal- ities
1985	6.1	4	197	1996	7.9	3	342
1986	6.4	2	5	1997	9.9	3	3
1987	6.6	4	231	1998	10.5	1	1
1988	6.7	3	285	1999	10.9	2	12
1989	6.6	11	278	2000	11.1	2	89
1990	7.8	6	39	2001	10.6	6	531
1991	7.5	4	62	2002	10.3	0	0
1992	7.5	4	33	2003	10.2	2	22
1993	7.7	1	1	2004	10.8	1	13
1994	7.8	4	239	2005	10.9	3	22
1995	8.1	2	166	2006	11.2	2	50

Source: National Transportation Safety Board.

Exercices

- Sample mean: ≈ 3.18 accidents per year
- Sample median: 3 accidents per year
- Sample mode: 2 accidents per year (most frequent value)
- Sample standard deviation: ≈ 2.31 accidents

U.S. Airline Safety, Scheduled Commercial Carriers, 1985–2006

Year	Departures (millions)	Acci- dents	Fatal- ities	Year	Departures (millions)	Acci- dents	Fatal- ities
1985	6.1	4	197	1996	7.9	3	342
1986	6.4	2	5	1997	9.9	3	3
1987	6.6	4	231	1998	10.5	1	1
1988	6.7	3	285	1999	10.9	2	12
1989	6.6	11	278	2000	11.1	2	89
1990	7.8	6	39	2001	10.6	6	531
1991	7.5	4	62	2002	10.3	0	0
1992	7.5	4	33	2003	10.2	2	22
1993	7.7	1	1	2004	10.8	1	13
1994	7.8	4	239	2005	10.9	3	22
1995	8.1	2	166	2006	11.2	2	50

Source: National Transportation Safety Board.

Exercices

1)

- Identify the characteristics under study.
- Specify whether they are discrete or continuous variables

2)

- Compute the proportion of employees in the age groups ≤ 40 years.
- Compute the same proportions for men and women separately.
- Interpret the results.

Age Range

Men

Women

< 20 years

32

51

20–30

1309

2118

30–40

1902

3025

40–50

1730

2330

50–60

1468

1624

> 60 years

114

131

Exercices

1)

- Age of employees → Quantitative continuous variable.
Gender (male/female) → Qualitative nominal variable

2)

- Men: 49.5%
- Women: 56.0%
- All employees 53.3%.

Age Range	Men	Women
< 20 years	32	51
20–30	1309	2118
30–40	1902	3025
40–50	1730	2330
50–60	1468	1624
> 60 years	114	131

Exercices

3)

Compute for men and women :

- Mean age
- Median
- Standard deviation

Age Range

Men

Women

< 20 years

32

51

20–30

1309

2118

30–40

1902

3025

40–50

1730

2330

50–60

1468

1624

> 60 years

114

131

Exercices

3)

- Mean age (Men) = 39.7 years
- Mean age (Women) = 38.4 years
- Median = 38 years (both sexes)
- Standard deviation (Men) = 10.5,
(Women) = 10.1

Age Range	Men	Women
< 20 years	32	51
20–30	1309	2118
30–40	1902	3025
40–50	1730	2330
50–60	1468	1624
> 60 years	114	131

Lab Python
